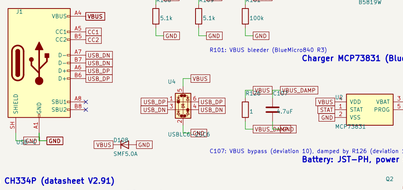
Lira: nRF52840 (Ebyte E73-2G4M08S1C), USB-C + Li-Po, 57 keys, built-in USB sound card

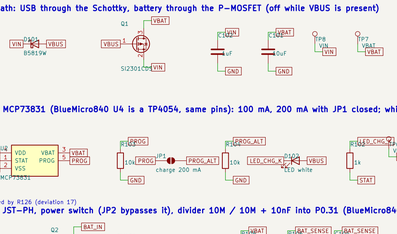
Power / USB / programming copied from the BlueMicro840 (jpconstantineau/NRF52-Board), a board built with this module, assembled by JLCPCB and sold. Differences from it:

1. Second regulator (switched 3.3 V for external devices, P0.12 enable) omitted: nothing external to power.
2. USBLC6-2SC6 ESD array added on D+ / D− / VBUS at the USB-C receptacle.
3. Power switch SW101 (C&K JS1020115AQN, 300 mA) in the battery + line, as on ZMK keyboards; JP2 bypasses it.
4. JP1 + R105: closing the solder jumper puts a second 10k on PROG: charge current 100 mA -> 200 mA.
5. Reset push button SW102 to GND (the BlueMicro840 has only a test point).
6. USB-C shield pads to GND (left open on the BlueMicro840).
7. LEDs: charging LED white, status LED red on P1.04 only (BlueMicro840: green charging, red P1.04, blue P1.10).
8. DCCH inductor pads (DNP on the BlueMicro840) not fitted at all: P1.10 is a free GPIO here, P1.02 switches the sound card.
9. Test points added on VBUS, VBAT and VIN (BlueMicro840: SWDIO, SWCLK, VCC, GND, RESET, P1.02).
10. C107 4.7 uF added on VBUS at the charger's VDD pin (the MCP73831 datasheet asks for 4.7 uF; Nordic's reference circuit has 4.7 uF on VBUS; the BlueMicro840 has none).
11. C108 1 uF at the regulator output and C109 100 nF at the module: the 4.7 uF C103 sits at the module's supply pins (VDD / VDDH), 25 mm away (the module has about 12 uF inside; the TPS7A02 allows 1–22 uF on its output). C102 on VIN is 1 uF instead of 10 uF: together they keep the capacitance charged at USB attach near USB 2.0's 10 uF.
12. Charger MCP73831 instead of the TP4054, same pins and same PROG resistor (1000 V / R_PROG): it is stable with no battery, so the board also works when built without one (its STAT pin, and the white LED, stay off).
13. Battery divider 10M + 10M with 10 nF (BlueMicro840: 806k + 2M, 1 nF): the MCP73831 recognises a missing battery only if VBAT sees more than 7 MOhm (datasheet 4.2): 10 nF >= 2*12 x the SAADC's 2.5 pF keeps the reading within 1 LSB (Nordic, 'Measuring Lithium battery voltage with nRF52'), and RMK samples it every 30 s, Drain 0.2 uA.
14. Built-in USB sound card: the USB-C data lines go (through the ESD array) to a CH334P hub; hub port 3 goes to the module's D+ / D−, port 2 to a CM108B codec (C-Media demo board v1.33) with a 3.5 mm headset jack.
15. Reverse-battery P-MOSFET Q2 (Si2301) between the battery connector and the switch, gate to GND: a cell plugged in the wrong way round finds it off (its body diode blocks), so the charger's VBAT pin never sees a negative voltage; the right way round it is on (Vgs = −Vbat) and charging goes through it.
16. Regulator TPS7A0233 (25 nA quiescent, 200 mA) instead of the AP2112K (55 uA): same SOT-23-5 pins (IN, GND, EN, NC, OUT); the 3.3 V rail needs < 50 mA (module, status LED).
17. VBUS: an SMF5.0A TVS at the receptacle, and C107 (4.7 uF) through 1 ohm: a ceramic-only VBUS rings on hot plug to 6–8 V through the cable's inductance (simulated, 0.5–1 uH); damped, 5.0–6.6 V, without adding capacitance.

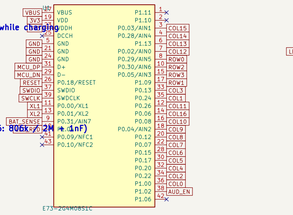
USB-C (sink: 5.1k on each CC) and ESD protection



Power path: USB through the Schottky, battery through the P-MOSFET (off while VBUS is present)

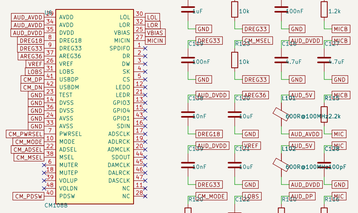


nRF52840 module, normal-voltage mode (VDD = VDDH = 3.3 V); reset button; status LED on P1.04 (bootloader LED); SWD pads



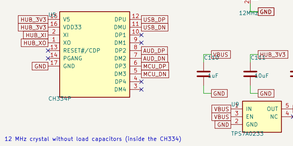
Sound card CM108B (C-Media demo board v1.33 sheet 4, app notes v1.50); headset jack, CTIA

Open pins: internal pulls select headset mode (MODE low), 100 mA bus power (PWRSEL high).
Internal ADC (ADSEL low), mixer on (MSEL high): R120–R125 strap them (10k). Jack: tip L, ring 1 R, ring 2 GND, sleeve mic.

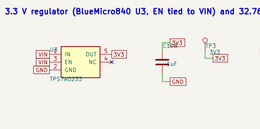

CM108B
Sound card supply: TPS22919 load switch, on only in USB mode (P1.02); headphone amplifier TPA6132A2 (DirectPath, 0 dB; −6 dB with JP3 closed); headset jack, CTIA
TPS22919
TPA6132A2
PJ-3200-4A
PESD5V0
Differences from the C-Media demo board:
1. Supply: VBUS through a TPS22919 load switch (soft start 1.3 ms, output discharge) to AUD_5V. ON = P1.02 with a 100k pull-downs: the sound card is powered only in USB mode (firmware src/audio_gate.rs). A hard switch-on would dip VBUS enough to reset the hub (CH334 datasheet 6.1).
2. Headphones: a TPA6132A2 DirectPath amplifier after LOL / LOR (demo board: 100 uF + 39R straight to the jack): about 0 ohm output impedance, no coupling capacitors, 0 dB gain (−6 dB with JP3 closed, for very sensitive earphones), inputs single-ended through 1 uF (8 Hz); EN about 50 ms after the supply (1M + 100 nF from P1.02, into the pin through 33k: EN may not exceed VDD + 0.3 V, and C137 still holds 3.3 V for a moment when the cable is pulled) and off at once through a Schottky. The switch's output discharge goes through 1k (TPS22919 9.2.2.2): the 5 V falls in about 3 ms, so the amplifier is off before the codec loses its supply (TPA6132A2 p.12, no pop).
3. Amplifier off in USB suspend: the codec's PDSW (open drain, low in operation, open in suspend; app note 1.50) with a 100k pull-up to AUD_5V drives a 2N7002 that holds the amplifier's EN low while the host sleeps (2 mA -> < 1 uA); the codec itself stays enumerated and draws its suspend current.
4. Hub supply: V5 and VDD33 both on a 3.3 V regulator (TPS7A0233) fed from VBUS (CH334 datasheet 6.4), instead of V5 on VBUS: V5's absolute maximum is 5.5 V and a USB hot plug rings above it; the regulator's input is rated 6.5 V. The regulator is on only with VBUS (the hub draws nothing from the battery).
5. Microphone: bias 1.2k + 2.2k (demo board 2.2k + 8.2k, for PC microphones) for phone headsets (Android: 1.8–2.9 V through 2.2k); 1k in series before MICIN against the step when a plug goes in.
6. The demo board's two 10 nF capacitors DVDD – DREG18 and DVDD – DREG33 ('close to IC', for ESD) are fitted.
7. The amplifier's SGND (pin 15) is grounded at the amplifier (its ground pad and a via into the top ground plane), not on a track of its own to the jack's ground pin as TI suggests (datasheet 10.3): with both layers poured the difference is a few millivolts.
8. The four mode jumpers are also strapped with 10k (MODE/ADSEL low, PWRSEL/MSEL high to DREG33) on top of the chip's internal pulls. D105–D107 are bidirectional ESD clamps (PESD5V0S1BA) on the jack's left, right and microphone contacts: a DirectPath output swings below GND, so the clamp must be bidirectional.

USB hub CH334P (datasheet V2.91)

upstream from the receptacle, port 3 -> module, port 2 -> codec
V5 = VDD33 = 3.3 V from U9, only with VBUS (datasheet 6.4)



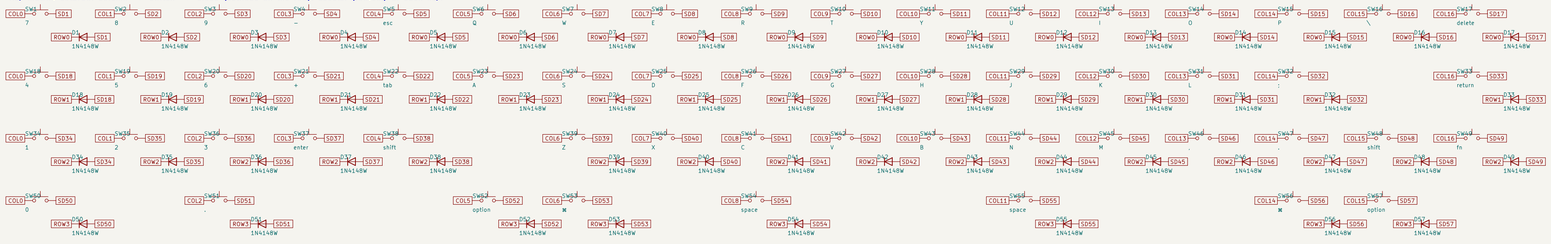
3.3 V regulator (BlueMicro840 U3, EN tied to VIN) and 32.768 kHz crystal (BlueMicro840 Y2 C1 C2)



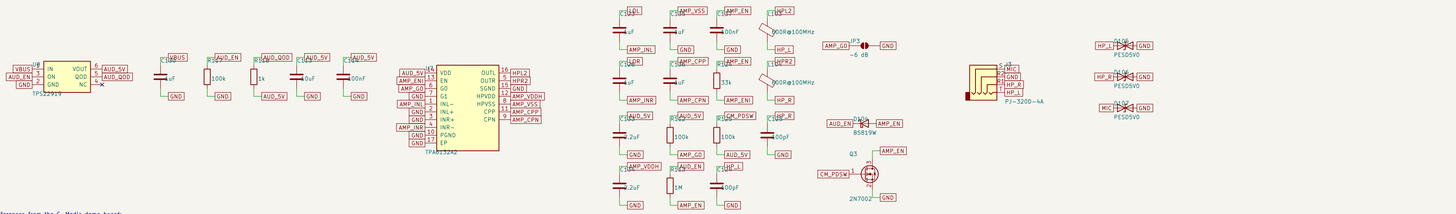
Supplies: VBUS from the receptacle, BAT+ from the cell, VIN from the power path, codec supplies from VBUS through ferrite beads



Key matrix, col2row: column -> switch -> diode (cathode on the row). Labels: (row,col) of each key.



Sound card supply: TPS22919 load switch, on only in USB mode (P1.02); headphone amplifier TPA6132A2 (DirectPath, 0 dB; −6 dB with JP3 closed); headset jack, CTIA



Differences from the C-Media demo board:

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